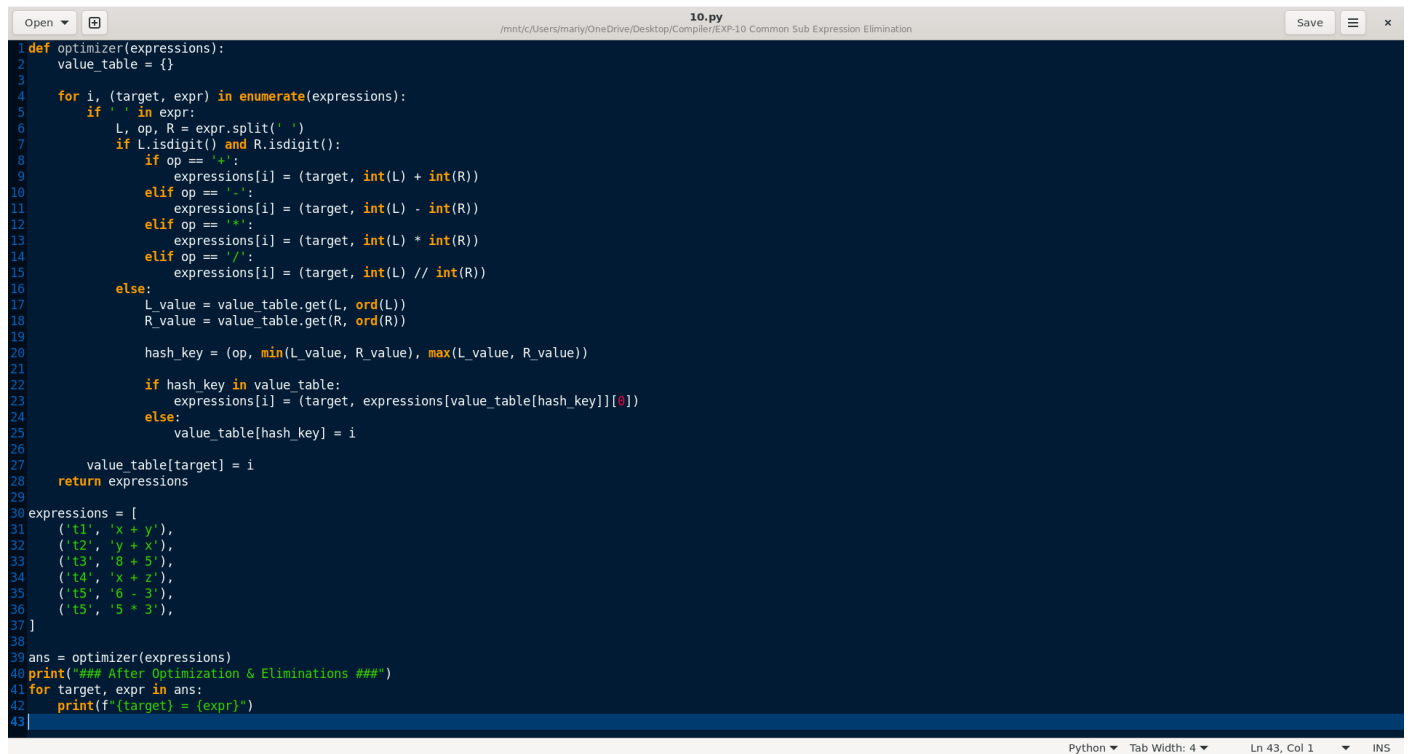


EXP – 10 Common Sub Expression Elimination

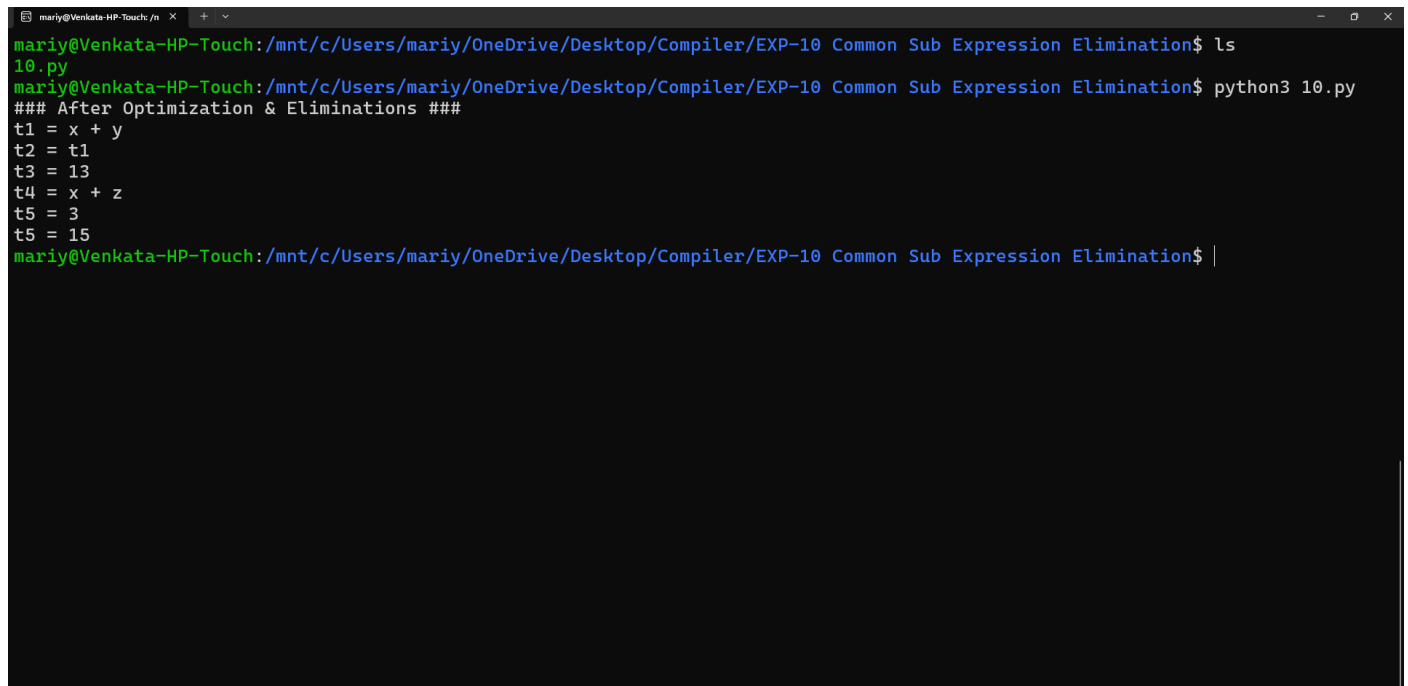
Code:



```
1 def optimizer(expressions):
2     value_table = {}
3
4     for i, (target, expr) in enumerate(expressions):
5         if ' ' in expr:
6             L, op, R = expr.split(' ')
7             if L.isdigit() and R.isdigit():
8                 if op == '+':
9                     expressions[i] = (target, int(L) + int(R))
10                elif op == '-':
11                    expressions[i] = (target, int(L) - int(R))
12                elif op == '*':
13                    expressions[i] = (target, int(L) * int(R))
14                elif op == '/':
15                    expressions[i] = (target, int(L) // int(R))
16            else:
17                L_value = value_table.get(L, ord(L))
18                R_value = value_table.get(R, ord(R))
19
20                hash_key = (op, min(L_value, R_value), max(L_value, R_value))
21
22                if hash_key in value_table:
23                    expressions[i] = (target, expressions[value_table[hash_key]][0])
24                else:
25                    value_table[hash_key] = i
26
27            value_table[target] = i
28    return expressions
29
30 expressions = [
31     ('t1', 'x + y'),
32     ('t2', 'y + x'),
33     ('t3', '8 + 5'),
34     ('t4', 'x + z'),
35     ('t5', '6 - 3'),
36     ('t5', '5 + 3'),
37 ]
38
39 ans = optimizer(expressions)
40 print("### After Optimization & Eliminations ###")
41 for target, expr in ans:
42     print(f"{target} = {expr}")
43
```

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Output:



```
mariy@Venkata-HP-Touch: /mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-10 Common Sub Expression Elimination$ ls
10.py
mariy@Venkata-HP-Touch: /mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-10 Common Sub Expression Elimination$ python3 10.py
### After Optimization & Eliminations ###
t1 = x + y
t2 = t1
t3 = 13
t4 = x + z
t5 = 3
t5 = 15
mariy@Venkata-HP-Touch: /mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-10 Common Sub Expression Elimination$
```